

● EASY

● ALGEBRA

● ANSWER KEY

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# Systems of two linear equations in two variables

01

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**Q1****23**

The correct answer is 23. Since it's given that  $x = 2$ , the value of  $y$  can be found by substituting 2 for  $x$  in the first equation and solving for  $y$ . Substituting 2 for  $x$  yields  $3(2) + y = 29$ , or  $6 + y = 29$ . Subtracting 6 from both sides of this equation yields  $y = 23$ .

**Q2****B****B. 3**

Choice B is correct. Let  $n$  be the number of novels and  $m$  be the number of magazines that Sadie purchased. If Sadie purchased a total of 11 novels and magazines, then  $n + m = 11$ . It is given that the combined price of 11 novels and magazines is \$20. Since each novel sells for \$4 and each magazine sells for \$1, it follows that  $4n + m = 20$ . So the system of equations below must hold.

$$\begin{aligned}4n + m &= 20 \\ n + m &= 11\end{aligned}$$

Subtracting corresponding sides of the second equation from the first equation yields  $3n = 9$ , so  $n = 3$ . Therefore, Sadie purchased 3 novels.

Choice A is incorrect. If 2 novels were purchased, then a total of \$8 was spent on novels. That leaves \$12 to be spent on magazines, which means that 12 magazines would have been purchased. However, Sadie purchased a total of 11 novels and magazines. Choices C and D are incorrect. If 4 novels were purchased, then a total of \$16 was spent on novels. That leaves \$4 to be spent on magazines, which means that 4 magazines would have been purchased. By the same logic, if Sadie purchased 5 novels, she would have no money at all (\$0) to buy magazines. However, Sadie purchased a total of 11 novels and magazines.

**Q3****B****B.  $(2, 4)$** 

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Choice B is correct. A solution to a system of equations must be the solution to each equation in the system. It follows that if  $x, y$  is a solution to the system, the point  $x, y$  lies on the graph in the  $xy$ -plane of each equation in the system. The point that lies on each graph of the system of linear equations shown is their intersection point  $2, 4$ . Therefore, the solution to the system is  $2, 4$ .

Choice A is incorrect. The point  $0, 5$  lies on one, but not both, of the graphs of the linear equations shown.

Choice C is incorrect. The point  $5, 10$  lies on one, but not both, of the graphs of the linear equations shown.

Choice D is incorrect. The point  $10, 0$  lies on one, but not both, of the graphs of the linear equations shown.

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## Q4

B

B.  $f + r = 214$

$$11f + 8.25r = 2,145$$

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Choice B is correct. The movie theater sells  $f$  full-price tickets and  $r$  reduced-price tickets, so the total number of tickets sold is  $f + r$ . Since the movie theater sold a total of 214 full-price and reduced-price tickets for one movie showing, it follows that  $f + r = 214$ . The movie theater charges \$11 for each full-price ticket; thus, the sales for full-price tickets, in dollars, is given by  $11f$ . The movie theater charges \$8.25 for each reduced-price ticket; thus, the sales for reduced-price tickets, in dollars, is given by  $8.25r$ . Therefore, the total sales, in dollars, for the movie showing is given by  $11f + 8.25r$ . Since the total sales for all full-price and reduced-price tickets is \$2,145, it follows that  $11f + 8.25r = 2,145$ .

Choice A is incorrect. This system of equations suggests that the movie theater sold a total of 2,145 full-price and reduced-price tickets for a total of \$214. Choice C is incorrect. This system suggests that the movie theater charges \$8.25 for each full-price ticket and \$11 for each reduced-price ticket. Choice D is incorrect. This system suggests that the movie theater charges \$8.25 for each full-price ticket and \$11 for each reduced-price ticket and sold a total of 2,145 tickets for a total of \$214.

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**Q5****A****A.** 15, 3

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Choice A is correct. The second equation in the given system defines the value of  $x$  as  $5y$ . Substituting  $5y$  for  $x$  into the first equation yields  $5y + y = 18$  or  $6y = 18$ . Dividing each side of this equation by 6 yields  $y = 3$ . Substituting 3 for  $y$  in the second equation yields  $5(3) = x$  or  $x = 15$ . Therefore, the solution  $x, y$  to the given system of equations is 15, 3.

Choice B is incorrect. Substituting 16 for  $x$  and 2 for  $y$  in the second equation yields  $5(2) = 16$ , which is not true. Therefore, 16, 2 is not a solution to the given system of equations.

Choice C is incorrect. Substituting 17 for  $x$  and 1 for  $y$  in the second equation yields  $5(1) = 17$ , which is not true. Therefore, 17, 1 is not a solution to the given system of equations.

Choice D is incorrect. Substituting 18 for  $x$  and 0 for  $y$  in the second equation yields  $5(0) = 18$ , which is not true. Therefore, 18, 0 is not a solution to the given system of equations.

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**Q6****30**

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The correct answer is 30. It's given that  $x + y = 125$ . Substituting 125 for  $x + y$  in the second equation of the given system yields  $125 + y = 155$ . Subtracting 125 from both sides of this equation yields  $y = 30$ . Therefore, the value of  $y$  is 30.

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**Q7****B**

B.  $20c + 10s = 700$   
 $25c + 12s = 850$

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Choice B is correct. The number of coins collected can be multiplied by  $c$  to give the score from the points earned from coins. Similarly, the number of stars collected can be multiplied by  $s$  to give the score from the points earned from the stars. Therefore, the total score each time Angela played is  $20c + 10s = 700$ , and the total score the second time she played is  $25c + 12s = 850$ .

Choices A, C, and D are incorrect and may result from misidentifying the terms of the equation. Choice A switches coins and stars, choice C switches stars and points, and choice D misidentifies coins, stars, and points.

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**Q8****4**

The correct answer is 4. A solution to a system of equations must satisfy each equation in the system. It follows that if  $x, y$  is a solution to the system, the point  $x, y$  lies on the graph in the  $xy$ -plane of each equation in the system. According to the graph, the point  $x, y$  that lies on the graph of each equation in the system is 4, 1. Therefore, the solution to the system is 4, 1. It follows that the value of  $x$  is 4.

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**Q9****C****C.** 36

Choice C is correct. A system of two linear equations has infinitely many solutions if one equation is equivalent to the other. This means that when the two equations are written in the same form, each coefficient or constant in one equation is equal to the corresponding coefficient or constant in the other equation multiplied by the same number. The equations in the given system of equations are written in the same form, with  $x$  and  $y$  on the left-hand side of the equation and a constant on the right-hand side of the equation. The coefficients of  $x$  and  $y$  in the second equation are equal to the coefficients of  $x$  and  $y$ , respectively, in the first equation multiplied by 4:  $8 = 2(4)$  and  $28 = 7(4)$ . Therefore, the constant in the second equation must be equal to 4 times the constant in the first equation:  $a = 9(4)$ , or  $a = 36$ .

Choices A, B, and D are incorrect. When  $a = 4$ ,  $a = 9$ , or  $a = 54$ , the given system of equations has no solution.

**Q10****C****C.** 15

Choice C is correct. The given system of linear equations can be solved by the substitution method. Substituting  $-3x$  for  $y$  from the first equation in the given system into the second equation yields  $4x + -3x = 15$ , or  $x = 15$ .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the absolute value of  $y$ , not the value of  $x$ .

**Notes**

Accept equivalent forms (e.g.  $0.25 \equiv 1/4$ ). For grid-in items, decimal and fraction answers are both valid.

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